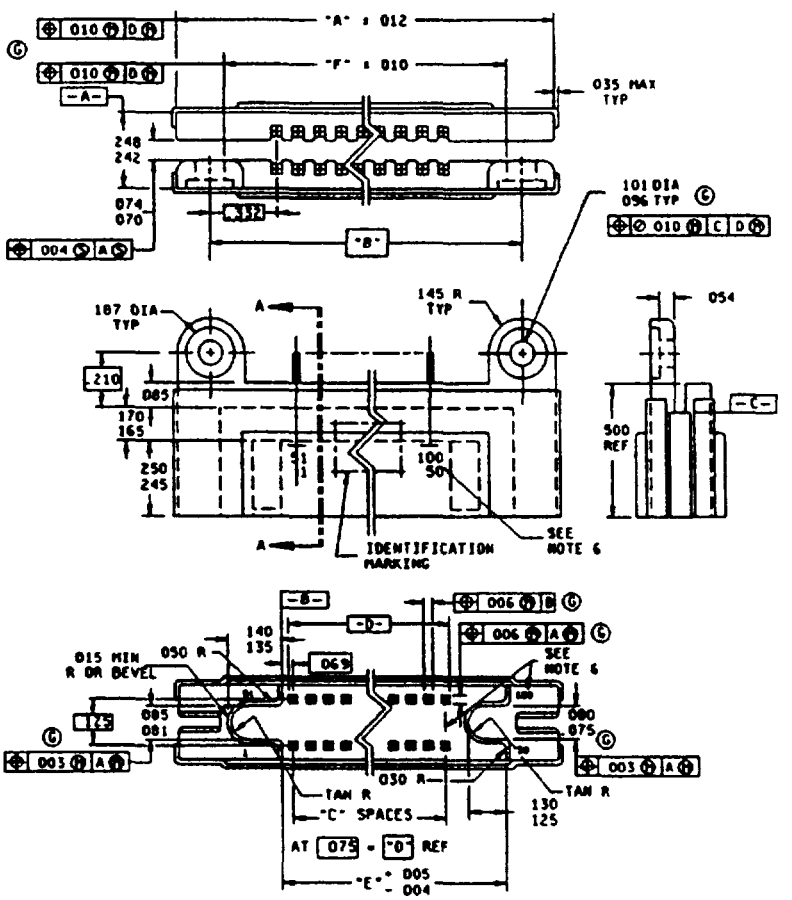
| INCH-POUND |
MIL-C-55302/23G
7 June 1993
SUPERSEDING
MIL-C-55302/23F
31 October 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES PLUG, DECADE INCREMENT 10 THRU 120, 150, 160, AND 180 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (075 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-55302.



STYLE 1 (INTERNAL KEYING)

(G) FIGURE 1. Connectors, plug, 075 (1.90 mm) spacing

(G) denotes changes

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

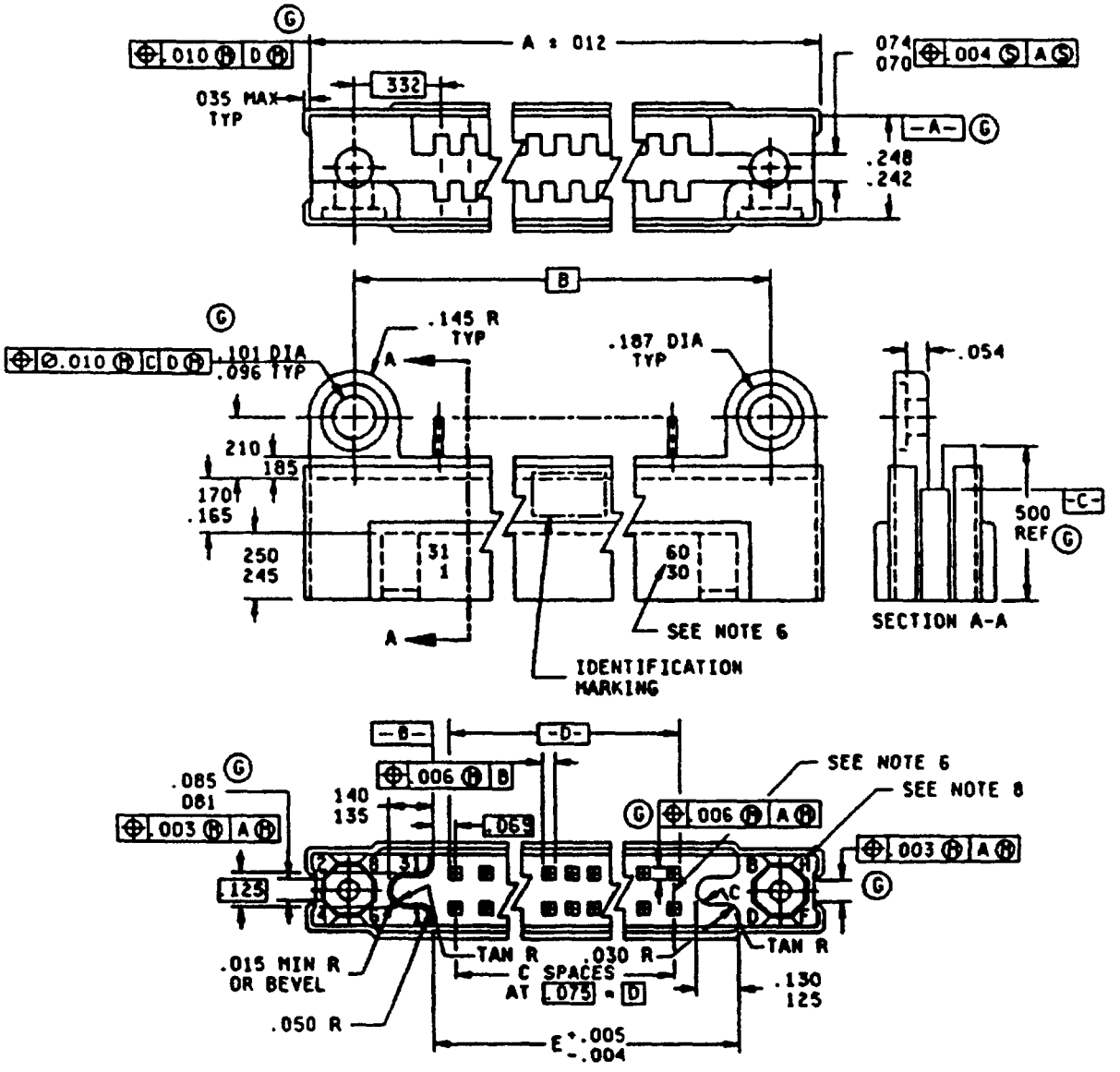
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Dash 1/ numbers	Number of contacts	Dimensions						Contact identification numbers
		A	B	C	D	E	F	
01	10	1 255 (31.88)	965 (24 51)	4	300 (7.62)	575 (14 61)	855 (21.72)	1 - 5 6 - 10
02	20	1 630 (41 40)	1 340 (34 04)	9	.675 (17 15)	950 (24 13)	1.230 (31.24)	1 - 10 11 - 20
03	30	2 005 (50 93)	1 715 (43 56)	14	1.050 (26 67)	1 325 (33 66)	1.605 (40 77)	1 - 15 16 - 30
04	40	2 380 (60.45)	2 090 (53 09)	19	1 425 (36 20)	1 700 (43.18)	1 980 (50 29)	1 - 20 21 - 40
05	50	2 755 (69.98)	2 465 (62.61)	24	1 800 (45 72)	2 075 (52 71)	2.355 (59.82)	1 - 25 26 - 50
06	60	3 130 (79 50)	2 840 (72 14)	29	2 175 (55 25)	2 450 (62 23)	2 730 (69.34)	1 - 30 31 - 60
07	70	3 505 (89 03)	3 215 (81 66)	34	2 550 (64.77)	2 825 (71 76)	3.105 (78 87)	1 - 35 36 - 70
08	80	3 880 (98.55)	3 590 (91 19)	39	2 925 (74.30)	3.200 (81.28)	3.480 (88 39)	1 - 40 41 - 80
09	90	4 255 (108 08)	3 965 (100 71)	44	3.300 (83 82)	3.575 (90 81)	3.855 (97.92)	1 - 45 46 - 90
10	100	4.630 (117.60)	4 340 (110 24)	49	3.675 (93.35)	3 950 (100.33)	4 230 (107.44)	1 - 50 51 - 100
11	110	5 005 (127 13)	4.715 (119.76)	54	4.050 (102.87)	4 325 (109.86)	4.605 (116.97)	1 - 55 56 - 110
12	120	5.380 (136.65)	5.090 (129 29)	59	4.425 (112.40)	4.700 (119.38)	4.980 (126 49)	1 - 60 61 - 120
13	150	6 505 (165 23)	6 215 (157.86)	74	5.550 (140.97)	5.825 (147.96)	6.150 (155.07)	1 - 75 76 - 150
14	160	6.880 (174 75)	6 590 (167 39)	79	5.925 (150 50)	6.200 (157.48)	6 480 (164.59)	1 - 80 81 - 160
15	180	7 630 (193 80)	7 340 (186 44)	89	6.675 (169.55)	6 950 (176.53)	7.230 (183.64)	1 - 90 91 - 180

1/ Dash number 01 through 15 are for style 1 (internal keying) connectors

(G) FIGURE 1 Connectors, plug, 075 (1 90 mm) spacing - Continued

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STYLE 2 (EXTERNAL KEYING)

(G) FIGURE 1. Connectors, plug, .075 (1.90 mm) spacing - Continued.

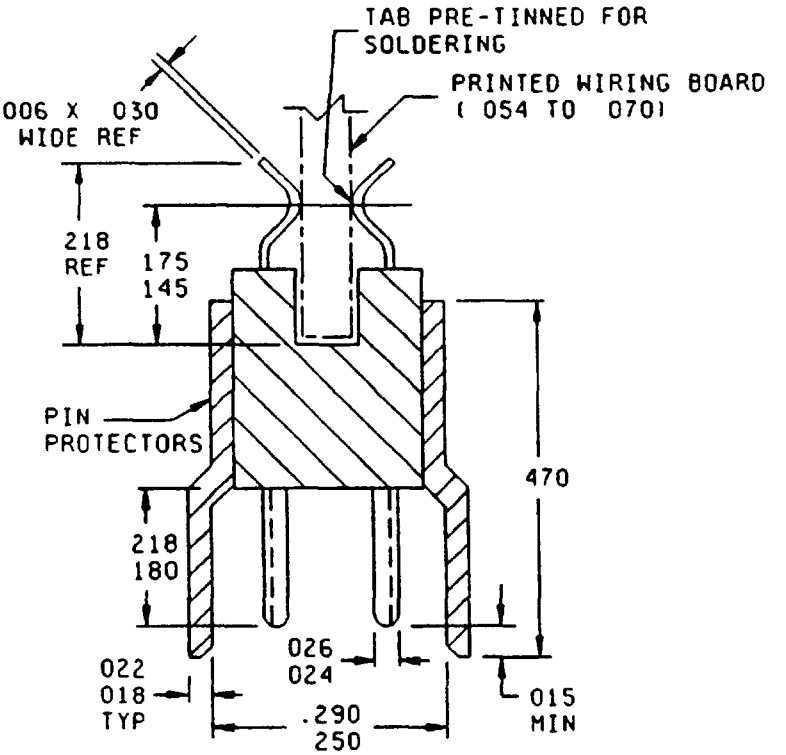
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Dash 1/ numbers	Number of contacts	Dimensions					Contact identification numbers
		A	B	C	D	E	
16	10	1 255 (31 88)	.965 (24 51)	4	.300 (7 62)	.575 (14 61)	1 - 5 6 - 10
17	20	1 630 (41 40)	1 340 (34 04)	9	675 (17 15)	950 (24 13)	1 - 10 11 - 20
18	30	2 005 (50 93)	1 715 (43 56)	14	1 050 (26 67)	1 325 (33 66)	1 - 15 16 - 30
19	40	2 380 (60 45)	2 090 (53 09)	19	1 425 (36 20)	1 700 (43.18)	1 - 20 21 - 40
20	50	2 755 (69 98)	2 465 (62 61)	24	1.800 (45.72)	2.075 (52 71)	1 - 25 26 - 50
21	60	3 130 (79.50)	2 840 (72.14)	29	2 175 (55.25)	2.450 (62.23)	1 - 30 31 - 60
22	70	3 505 (89 03)	3 215 (81 66)	34	2.550 (64.77)	2.825 (71 76)	1 - 35 36 - 70
23	80	3 880 (98 55)	3 590 (91 19)	39	2 925 (74 30)	3 200 (81 28)	1 - 40 41 - 80
24	90	4 255 (108 08)	3 965 (100 71)	44	3 300 (83 82)	3 575 (90.81)	1 - 45 46 - 90
25	100	4 630 (117 60)	4 340 (110 24)	49	3 657 (93 35)	3 950 (100 33)	1 - 50 51 - 100
26	110	5 005 (127.13)	4.715 (119 76)	54	4.050 (102.87)	4 325 (109.86)	1 - 55 56 - 110
27	120	5 380 (136 65)	5.090 (129.29)	59	4 425 (112 40)	4 700 (119.38)	1 - 60 61 - 120
28	150	6 505 (165 23)	6 215 (157 86)	74	5.550 (140 97)	5 825 (147 96)	1 - 75 76 - 150
29	160	6 880 (174 75)	6 590 (167 39)	79	5 925 (150 50)	6 200 (157.48)	1 - 80 81 - 160
30	180	7 630 (193 80)	7 340 (186 44)	89	6 675 (169 55)	6 950 (176.53)	1 - 90 91 - 180

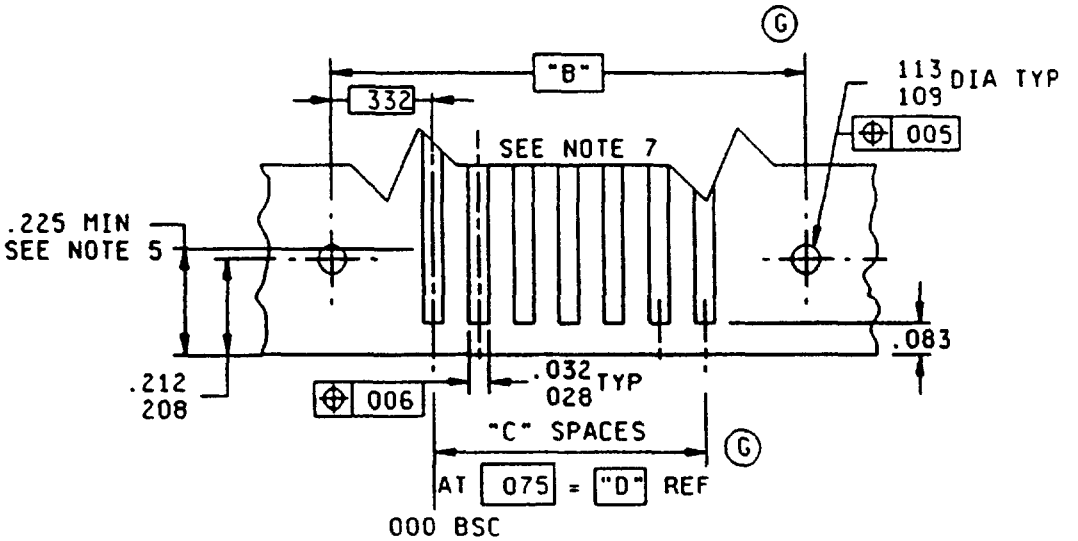
1/ Dash number 16 through 30 are for style 2 (external keying) connectors

(G) FIGURE 1 Connectors, plug, 075 (1 90 mm) spacing - Continued

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SECTION A-A



RECOMMENDED PRINT WIRING BOARD FOR STYLES 1 AND 2

(G) FIGURE 1 Connectors, plug, 075 (1.90 mm) spacing - Continued

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Inches	mm	Inches	mm	Inches	mm
003	0 08	.069	1.75	170	4 32
004	0 10	070	1 78	175	4 44
005	0 13	074	1 88	180	4 57
006	0 15	075	1 90	187	4 75
008	0 20	080	2 03	208	5 28
010	0 25	081	2 06	210	5 33
012	0 30	083	2 11	212	5 38
015	0 38	085	2.16	215	5 46
018	0 46	.096	2.44	.218	5.54
022	0 56	101	2.57	.225	5 72
024	0 61	.109	2 77	.242	6 15
026	0 66	.113	2 87	245	6 22
028	0 71	125	3 18	.248	6 30
030	0 76	130	3 30	250	6 35
032	0 81	135	3.43	.290	7 37
035	0 89	140	3 56	332	8 43
050	1 27	145	3 68	.470	11 94
054	1 37	165	4 19	500	12 70

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles
4. These connectors mate with connectors specified in MIL-C-55302/24 or MIL-C-55302/25
5. Minimum pad length on printed circuit board.
6. Numbers indicating, every 5 cavities marked or molded on side Numbers indicating end cavities and lines indicating every 5 cavities molded or marked on mating face
7. Termination layout on .025 (0 64 mm) modular grid
8. Location indicators embossed on surface

(G) FIGURE 1 Connectors, plug, 075 (1 90 mm) spacing - Continued

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REQUIREMENTS

Design and construction

Dimensions and configuration See figure 1

Material

(G) Insulator body Insulator material shall be in accordance with MIL-C-55302 or MIL-M-24519 type GLCP-30F

(G) Contact material Contact material in accordance with MIL-C-55302 or Beryllium copper alloy CDA174 in accordance with ASTM B768

Plating: The contact plating for the engagement area, 150 inch minimum length, shall be gold per MIL-C-55302 over nickel per MIL-C-55302 The contact plating for the solder tail area, 160 inch minimum length, shall be tin lead over nickel, with a tin lead dip per MIL-C-55302 for 080 minimum length. The remainder of the contact shall be nickel plated per MIL-C-55302

Contact identification See figure 1

Coating The aluminum pin protector shall be coated per MIL-A-8625, type II or epoxy coated 001 to 003 inch thick

Contact rating 3.0 amperes maximum per contact, 2 25 amperes continuous per contact at room ambient

Keying See MIL-C-55302/31

Mating and unmating The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of 025 times the number of contacts and shall not exceed the measured insertion force

Contact resistance The average contact resistance of all contacts measured shall not exceed 015 ohm, and no individual contact pair shall have a resistance exceeding 020 ohm

Contact retention 3 pounds minimum

Dielectric withstanding voltage

Sea level 900 volts rms

High altitude 200 volts rms

Part number. M55302/23- (dash number from figure 1)

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 17

Review activities
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
DLA - ES

User activities.
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3858-05)

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